THREE EQUIVALENT CHARACTERIZATIONS OF UNRAMIFIEDNESS FOR A MORPHISM OF ALGEBRAIC STACKS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The characterization of DM morphisms through surjective etale covers by schemes or algebraic spaces. Invoked inside the counted proof (source lines 8362-8365 and 8378-8382). Produces the etale presentations used in conditions (2) and (3).
- The criterion for checking separatedness-type properties along a flat finitely presented cover. Invoked inside the counted proof (source lines 8368-8372). Reduces DM-ness of the morphism to that of a base change.
- The descent statement for unramifiedness along a flat finitely presented surjection. Invoked inside the counted proof (source lines 8388-8390) as a cross-chapter pointer. Allows the property to be checked after base change in the last step.
- The characterization of unramifiedness through a presentation and the definition of the property. Invoked near the end of the counted proof (source lines 8396-8400). Turns the unramifiedness of the presenting space into that of the morphism.

Source reference. Original definition definition-unramified: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8273–8273. Navigation pointer only: the counted text refers to the Stacks item definition-unramified, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4746–4746. Navigation pointer only: the counted

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text refers to the Stacks item lemma-DM, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-check-separated-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 727–727. Navigation pointer only: the counted text refers to the Stacks item lemma-check-separated-covering, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-compose-after-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 929–929. Navigation pointer only: the counted text refers to the Stacks item lemma-compose-after-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-etale-smooth-local-source-target: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7765–7765. Navigation pointer only: the counted text refers to the Stacks item lemma-etale-smooth-local-source-target, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference spaces-descent-lemma-descending-property-unramified: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8261–8261. Navigation pointer only: the counted text refers to the Stacks item spaces-descent-lemma-descending-property-unramified, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original theorem theorem-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4589–4589. Navigation pointer only: the counted text refers to the Stacks item theorem-DM, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. The following are equivalent*

1. *f is unramified,*
2. *f is DM and for any morphism $V \to \mathcal{Y}$ where V is an algebraic space and any étale morphism $U \to V \times_{\mathcal{Y}} \mathcal{X}$ where U is an algebraic space, the morphism $U \to V$ is unramified,*
3. *there exists some surjective, locally of finite presentation, and flat morphism $W \to \mathcal{Y}$ where W is an algebraic space and some surjective étale morphism $T \to W \times_{\mathcal{Y}} \mathcal{X}$ where T is an algebraic space such that the morphism $T \to W$ is unramified.*

Proof. Assume (1). Then f is DM and since being unramified is preserved by base change, we see that (2) holds.

Assume (2). Choose a scheme V and a surjective étale morphism $V \to \mathcal{Y}$. Choose a scheme U and a surjective étale morphism $U \to V \times_{\mathcal{Y}} \mathcal{X}$ (Lemma lemma-DM). Thus we see that (3) holds.

Assume $W \to \mathcal{Y}$ and $T \to W \times_{\mathcal{Y}} \mathcal{X}$ are as in (3). We first check f is DM. Namely, it suffices to check $W \times_{\mathcal{Y}} \mathcal{X} \to W$ is DM, see Lemma lemma-check-separated-covering. By Lemma lemma-compose-after-separated it suffices to check $W \times_{\mathcal{Y}} \mathcal{X}$ is DM. This follows from the existence of $T \to W \times_{\mathcal{Y}} \mathcal{X}$ by (the easy direction of) Theorem theorem-DM.

Assume f is DM and $W \to \mathcal{Y}$ and $T \to W \times_{\mathcal{Y}} \mathcal{X}$ are as in (3). Let V be an algebraic space, let $V \to \mathcal{Y}$ be surjective smooth, let U be an algebraic space, and let $U \to V \times_{\mathcal{Y}} \mathcal{X}$ is surjective and étale (Lemma lemma-DM). We have to check that $U \to V$ is unramified. It suffices to prove $U \times_{\mathcal{Y}} W \to V \times_{\mathcal{Y}} W$ is unramified by Descent on Spaces, Lemma spaces-descent-lemma-descending-property-unramified. We may replace $\mathcal{X}, \mathcal{Y}, W, T, U, V$ by $\mathcal{X} \times_{\mathcal{Y}} W, W, W, T, U \times_{\mathcal{Y}} W, V \times_{\mathcal{Y}} W$ (small detail omitted). Thus we may assume that $Y = \mathcal{Y}$ is an algebraic space, there exists an algebraic space T and a surjective étale morphism $T \to \mathcal{X}$ such that $T \to Y$ is unramified, and U and V are as before. In this case we know that

$$U \to V \text{ is unramified} \Leftrightarrow \mathcal{X} \to Y \text{ is unramified} \Leftrightarrow T \to Y \text{ is unramified}$$

by the equivalence of properties (1) and (2) of Lemma lemma-etale-smooth-local-source-target and Definition definition-unramified. This finishes the proof. □